



Chapter 2 NEW - good

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CHAPTER 2 : Numerical Differentiation & Integration

Differentiation

- Rules of differentiation
- Product rule
- Quotient rule
- Partial derivative
- Point of inflexion

2.1 Introduction

Differentiation is concerned with the rate of change of one quantity relative to another. If quantities are continuously changing, such as velocity, acceleration and current in a circuit, we use differentiation to study the nature of their change. Differentiation is also used to solve optimisation problems.

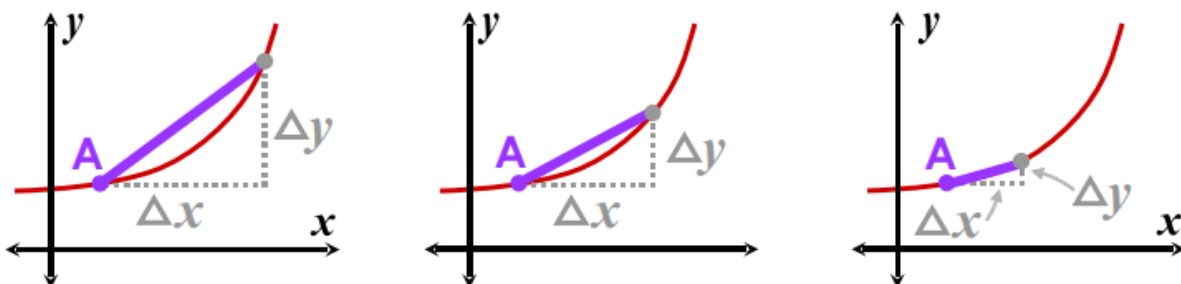
Differentiation is also known as derivative. In general, differentiation is finding a slope of a tangent line passing through a curve. Slope of the tangent is given as $\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$

Differentiation of function $y = f(x)$ is written as $\frac{dy}{dx} = f'(x)$ (read as f prime of x). The term of

$\frac{dy}{dx} = f'(x)$ is referring to the derivative of y with respect to x.

Estimating the Instantaneous Rate of Change

The diagram below show attempts to estimate the instantaneous gradient (the rate of change of y with respect to x)



Notice that the accuracy improves as Δx gets closer to zero. The instantaneous rate of change is written as :

$$\frac{dy}{dx} = \frac{\Delta y}{\Delta x} \text{ as } \Delta x \text{ approaches } 0.$$

2.2 Rules of differentiation

Finding the derivatives using the first principle is tedious and time consuming. However there are several technique based on the first principle that enable us to find the derivatives of a big group of functions more efficiently.

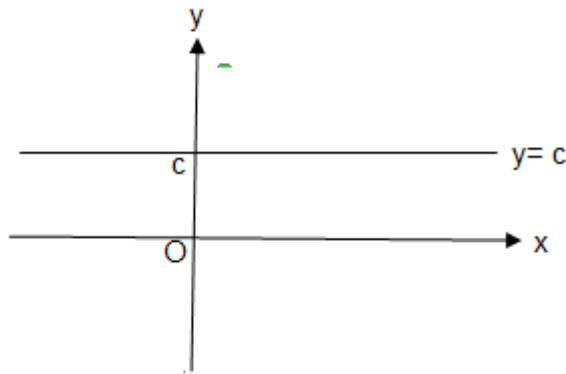
Definition of Derivatives

The derivatives of f at x is given by $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$ provided this limit exists.

The method of finding the derivative using the above definition is called the **first principle**.

Derivative of a constant

Consider the function $y = c$, where c is a constant. Geometrically, this function represents a straight line parallel to the x -axis, as shown in Figure below.



Since the gradient of the straight line $y = c$ is zero, therefore $\frac{dy}{dx} = 0$ or $\frac{d}{dx}(c) = 0$

Note that $\frac{dy}{dx}$ describes both **the rate of change** and **the gradient**.

2.2.1 Derivative of a Constant Function c

If c is any real number, then $\frac{d}{dx}(c) = 0$.

Example 1

Find $f'(x)$

- a. $f(x) = 10$, $f'(x) = 0$
- b. $f(x) = -5.1$, $f'(x) = \underline{\hspace{2cm}}$
- c. $f(x) = 1/5$, $f'(x) = \underline{\hspace{2cm}}$

2.2.2 Derivative of a Power Functions

The Power Rule

For any real number n, $\frac{d}{dx}(x^n) = nx^{n-1}$

and

$$\frac{d}{dx}(cx^n) = ncx^{n-1}, \text{ c is constant.}$$

Example 2

$f(x)$	$f'(x)$
$x = x^1$	1
x^2	$2x$
x^3	$3x^2$
$\vdots$	$\vdots$
x^n	nx^{n-1}

** How to Differentiate:

- **Multiply** by the power
- **Reduce** the power by one

Example 3

Find $\frac{dy}{dx}$

a. $y = x^5$ Solution : $\frac{dy}{dx} = 5x^4$	b. $y = \frac{1}{x^7} = x^{-7}$ Solution : $\frac{dy}{dx} = -7x^{-8} = -\frac{7}{x^8}$	c. $y = \sqrt{x}$ $= x^{1/2}$ Solution : $\frac{dy}{dx} = \frac{1}{2} x^{-1/2}$ $= \frac{1}{2\sqrt{x}}$
d. $y = 4x^{10}$ Solution: $\frac{dy}{dx} =$	e. $y = \frac{3}{x^4}$ Solution: $\frac{dy}{dx} =$	f. $y = \frac{1}{4\sqrt{x}}$ Solution: $\frac{dy}{dx} =$

Derivatives of Sums and Differences

The basic process of differentiation can be applied to every x-term in an algebraic expression.

If f and g are differentiable at x, then

$$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}[f(x)] \pm \frac{d}{dx}[g(x)] = f'(x) \pm g'(x)$$

$$y = ax^m + bx^n + \dots$$

$$\frac{dy}{dx} = amx^{m-1} + bnx^{n-1} + \dots$$

How to Differentiate:

- **Multiply** every x-term by the power
- **Reduce** the power of every x-term by one

**** Important**

Expressions **must be written** as the sum of individual terms before differentiating

Example 4

Differentiate each of the following functions.

a. $f(x) = 7 + x^2 + 3x^3$ Solution : $f(x) = \textcircled{7} + x^2 + 3x^3$ This disappears because $7 = 7x^0$ (multiply by zero) $f'(x) = 0 + 2x + 9x^2$ $f'(x) = 2x + 9x^2$	b. $g(x) = 3x^8 - 2x^5 + 6x + 1$ Solution : $g(x) = 3x^8 - 2x^5 + 6x + 1$ $g'(x) = 24x^7 - 10x^4 + 6 + 0$ $g'(x) = 24x^7 - 10x^4 + 6$	c. $p(x) = x^2 - \sqrt[5]{x}$ Solution:
d. $q(x) = \sqrt{x} + \frac{1}{\sqrt{x}}$ Solution:	e. $y = \frac{3}{\sqrt{x}} + 2\sqrt{x} + \frac{1}{x} - 5x + 7$ Solution:	

Product RuleIf $f(x) = u(x) \cdot v(x)$ and $u'(x)$ and $v'(x)$ both exist, then

$$\frac{d}{dx}[f(x)] = \frac{d}{dx}(u \cdot v) = uv' + vu'$$

Example 5: Find $f'(x)$

1) $f(x) = (2x + 7)(x^2 - 6x + 1)$ Solution: $u = 2x + 7$ $v = x^2 - 6x + 1$ $u' = 2$ $v' = 2x - 6$ By applying product rule : $f'(x) = uv' + vu'$ $= (2x + 7)(2x - 6) + (x^2 - 6x + 1)(2)$ $= 4x^2 - 12x + 14x - 42 + 2x^2 - 12x + 2$ $= 6x^2 - 10x - 40$	2) $f(x) = (3x^2 + 6)(2x - \frac{1}{4})$ Solution :
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Quotient Rule

If $f(x) = \frac{u(x)}{v(x)}$ and $u'(x)$ and $v'(x)$ both exist, and $v(x) \neq 0$ then

$$\frac{d}{dx}[f(x)] = \frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$$

Example 6:

1) $\frac{d}{dx}\left(\frac{x^2+1}{x^3+5}\right)$ Solution : $u = x^2 + 1$ $v = x^3 + 5$ $u' = 2x$ $v' = 3x^2$ By applying quotient rule: $\begin{aligned}\frac{d}{dx}\left(\frac{x^2+1}{x^3+5}\right) &= \frac{vu' - uv'}{v^2} \\ &= \frac{(x^3+5) \cdot 2x - (x^2+1)3x^2}{(x^3+5)^2} \\ &= \frac{2x^4 + 10x - 3x^4 - 3x^2}{(x^3+5)^2} \\ &= \frac{-x^4 - 3x^2 + 10x}{(x^3+5)^2}\end{aligned}$	2. $\frac{d}{dx}\left(\frac{2x^2-1}{x+5}\right)$ Solution :
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Exercise 2

Find $\frac{dy}{dx}$ for the following functions :

1. $y = 2x^2 - 5\sqrt{x}$	2. $y = \frac{1}{2}(x^4 + 7x + 1)$
3. $y = -\frac{1}{3}(x^7 + 2x - 9)$	4. $y = \sqrt{2}x + \frac{2}{\sqrt{x}}$
5. $y = \frac{1}{x^3} + \frac{1}{x^7}$	6. $y = \frac{1}{2\sqrt{x}} + 5\sqrt{x} - 8x + 9$
7. $y = (4x^2 - 1)(7x^3 + x)$	8. $y = (x^3 + 7x^2 - 8)\left(\frac{2}{x^3} + \frac{1}{x^4}\right)$
9. $g(x) = (x + 2)(2\sqrt{x} + x^2)$	10. $f(x) = (2x + 1)(1 - 4x^3)$

11. $h(x) = \frac{3x+5}{x+9}$	12. $p(x) = \frac{x^2+3}{x^2+5}$
13. $y = \frac{x}{x^2+1}$	14. $f(x) = \frac{\sqrt{x}}{x^2+2}$

Derivatives of trigonometric functions

1. $\frac{d}{dx}(\sin x) = \cos x$	7. $\frac{d}{dx}(\sin f(x)) = \cos f(x) \cdot f'(x)$
2. $\frac{d}{dx}(\cos x) = -\sin x$	8. $\frac{d}{dx}(\cos f(x)) = -\sin f(x) \cdot f'(x)$
3. $\frac{d}{dx}(\tan x) = \sec^2 x$	9. $\frac{d}{dx}(\tan f(x)) = \sec^2 f(x) \cdot f'(x)$
4. $\frac{d}{dx}(\sec x) = \sec x \tan x$	
5. $\frac{d}{dx}(\csc x) = -\csc x \cot x$	
6. $\frac{d}{dx}(\cot x) = -\csc^2 x$	

Derivatives of exponential functions

1. $\frac{d}{dx}(e^x) = e^x$	3. $\frac{d}{dx}(e^{f(x)}) = e^{f(x)} \cdot f'(x)$
2. $\frac{d}{dx}(a^x) = a^x \ln a$	4. $\frac{d}{dx}(a^{f(x)}) = a^{f(x)} \cdot f'(x) \ln a$

Example 7

Find $\frac{dy}{dx}$

- $y = e^{x^2}$
- $y = e^{\sin x}$
- $y = x^3 e^{4x}$

Solution:

a. $y = e^{x^2}$ $\frac{dy}{dx} = e^{x^2} \cdot \frac{d}{dx}(e^{x^2}) = e^{x^2} (2x)$ $= 2xe^{x^2}$	b. $y = e^{\sin x}$ $\frac{dy}{dx} = e^{\sin x} \cdot \frac{d}{dx}(\sin x) = e^{\sin x} (\cos x)$ $= \cos x e^{\sin x}$
c. $y = x^3 e^{4x}$	

Derivatives of Logarithmic Functions

1. $\frac{d}{dx}(\ln x) = \frac{1}{x}$	3. $\frac{d}{dx}(\ln f(x)) = \frac{1}{f(x)} \cdot f'(x)$
2. $\frac{d}{dx}(\log_b x) = \frac{1}{x \ln b}$	4. $\frac{d}{dx}(\log_b f(x)) = \frac{1}{f(x) \ln b} \cdot f'(x)$

Example 8

Differentiate the following functions.

- a. $y = \ln x^2$
- b. $f(x) = \ln(\sin x)$
- c. $h(x) = \ln(x \sin 2x)$
- d. $g(x) = \log_3(x)$

Solution:

a. $y = \ln x^2$	b. $f(x) = \ln(\sin x)$
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$\frac{dy}{dx} = \frac{1}{x^2} \bullet \frac{d}{dx}(x^2)$ $= \frac{1}{x^2} (2x)$ $= \frac{2}{x}$	
c. $h(x) = \ln(x \sin 2x)$	d. $g(x) = \log_3(x)$

Exercise 3.4:

1. Find $\frac{dy}{dx}$

a. $y = \frac{e^x + e^{-x}}{e^x - e^{-x}}$

ans : $-\frac{4}{(e^x - e^{-x})^2}$

b. $y = \frac{1 + e^x}{e^{2x}}$

ans : $\frac{-e^x - 2}{e^{2x}}$

c. $y = e^{x + \cos 2x}$

ans : $(1 - 2 \sin 2x)e^{x + \cos 2x}$

d. $y = \ln(\cos x)$

ans : $-\tan x$

e. $y = \ln(\ln x)$

ans : $\frac{1}{x \ln x}$

f. $y = \frac{e^x}{\ln x}$

ans : $\frac{xe^x \ln x - e^x}{x(\ln x)^2}$

2.2.3 Partial derivatives

Definition : If $z = f(x, y)$ and (x_0, y_0) is a point in the domain of f , then the **partial derivative of f with respect to x** at (x_0, y_0) [also called **the partial derivative of z with respect to x** at (x_0, y_0)] is the derivative at x_0 of the function that results when $y = y_0$ is held fixed and x is allowed to vary. This partial derivative is denoted by $f_x(x_0, y_0)$ and is given by

$$f_x(x_0, y_0) = \left. \frac{d}{dx} [f(x, y_0)] \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x}$$

Similarly, the **partial derivative of f with respect to y** at (x_0, y_0) [also called the **partial derivative of z with respect to y** at (x_0, y_0)] is the derivative at y_0 of the function that results when $x = x_0$ is held fixed and y is allowed to vary. This partial derivative is denoted by $f_y(x_0, y_0)$ and is given by

$$f_y(x_0, y_0) = \left. \frac{d}{dy} [f(x_0, y)] \right|_{y=y_0} = \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y}$$

Example 9: Find $f_x(1, 3)$ and $f_y(1, 3)$ for the function $f(x, y) = 2x^3y^2 + 2y + 4x$

Solution:

1.
$$f_x(x, y) = \frac{d}{dx} [2x^3y^2 + 2y + 4x] = 6x^2y^2 + 4$$

$$\begin{aligned} f_x(1, 3) &= 6(1)^2(3)^2 + 4 \\ &= 54 + 4 \\ &= 58 \end{aligned}$$

2.
$$f_y(x, y) = \frac{d}{dy} [2x^3y^2 + 2y + 4x] = 4x^3y + 2$$

$$\begin{aligned} f_y(1, 3) &= 4(1)^3(3) + 2 \\ &= 12 + 2 \\ &= 14 \end{aligned}$$

Example 10:

Let $f(x, y) = x^2y + 5y^3$. Find $f_x(1, 3)$ and $f_y(1, 3)$

Solution:

$$1. \quad f_x(x, y) = \frac{d}{dx}[x^2y + 5y^3] = 2xy$$

$$\begin{aligned} f_x(1, 3) &= 2(1)(3) \\ &= 6 \end{aligned}$$

$$2. \quad f_y(x, y) = \frac{d}{dy}[x^2y + 5y^3] = x^2 + 15y^2$$

$$\begin{aligned} f_y(1, 3) &= (1)^2 + 15(3)^2 \\ &= 1 + 135 \\ &= 136 \end{aligned}$$

Partial derivative notation

If $z = f(x, y)$, then the partial derivatives f_x and f_y are also denoted by the symbols

$$\frac{\partial f}{\partial x}, \quad \frac{\partial z}{\partial x}, \quad \text{and} \quad \frac{\partial f}{\partial y}, \quad \frac{\partial z}{\partial y}$$

Some typical notations for the partial derivatives of $z = f(x, y)$ at a point (x_0, y_0) are

$$\left. \frac{\partial f}{\partial x} \right|_{x=x_0, y=y_0}, \quad \left. \frac{\partial z}{\partial x} \right|_{(x_0, y_0)}, \quad \left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)}, \quad \frac{\partial f}{\partial x}(x_0, y_0), \quad \frac{\partial z}{\partial x}(x_0, y_0)$$

Example 11: Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if $z = x^4 \sin(xy^3)$

Solution:

$$\begin{aligned} 1) \quad \frac{\partial z}{\partial x} &= \frac{\partial}{\partial x}[x^4 \sin(xy^3)] = x^4 \frac{\partial}{\partial x}[\sin(xy^3)] + \sin(xy^3) \cdot \frac{\partial}{\partial x}(x^4) \\ &= x^4 \cos(xy^3) \cdot y^3 + \sin(xy^3) \cdot 4x^3 \\ &= x^4 y^3 \cos(xy^3) + 4x^3 \sin(xy^3) \end{aligned}$$

$$\begin{aligned}
 2) \quad \frac{\partial z}{\partial y} &= \frac{\partial}{\partial y}[x^4 \sin(xy^3)] = x^4 \frac{\partial}{\partial y}[\sin(xy^3)] + \sin(xy^3) \cdot \frac{\partial}{\partial y}(x^4) \\
 &= x^4 \cos(xy^3) \cdot 3xy^2 + \sin(xy^3) \cdot 0 \\
 &= 3x^5 y^2 \cos(xy^3)
 \end{aligned}$$

Exercise 3:

1. Let $f(x, y) = 3x^3 y^2$. Find

a) $f_x(x, y)$ b) $f_y(x, y)$ c) $f_x(1, 2)$ d) $f_y(-1, 5)$

2. Let $z = e^x \sin y$. Find

a) $\frac{\partial z}{\partial x}$ b) $\frac{\partial z}{\partial y}$ c) $\left. \frac{\partial z}{\partial x} \right|_{(2,3)}$ d) $\left. \frac{\partial z}{\partial y} \right|_{(0,3)}$

The Derivatives and Properties of a Graph

If $y = f(x)$ is a function which is differentiable in the interval (a,b) and $\frac{dy}{dx} = f'(x) > 0$

for all $x \in (a,b)$, then $y = f(x)$ is increasing in the interval (a,b) .

If $y = f(x)$ is a function which is differentiable in the interval (a,b) and $\frac{dy}{dx} = f'(x) < 0$

for all $x \in (a,b)$, then $y = f(x)$ is decreasing in the interval (a,b) .

Example:

Find the set of values of x for which the function

$$f(x) = \frac{2}{3}x^3 - \frac{1}{2}x^2 - 3x + 4 \text{ is}$$

a) increasing.

b) decreasing.

Solution:

$$f(x) = \frac{2}{3}x^3 - \frac{1}{2}x^2 - 3x + 4$$

$$f'(x) = 2x^2 - x - 3$$

When f is increasing,

$$f'(x) > 0$$

$$2x^2 - x - 3 > 0$$

$$(x+1)(2x-3) > 0$$

Use table:

Solution :

$$2x^2 - x - 3 > 0$$

Factoring, $(x+1)(2x-3) > 0$

The critical values involved are :

Note:
We want positive value

$$x + 1 = 0 \quad \text{and} \quad 2x - 3 = 0$$

$$x = -1 \qquad x = \frac{3}{2}$$

Plot them on a table :

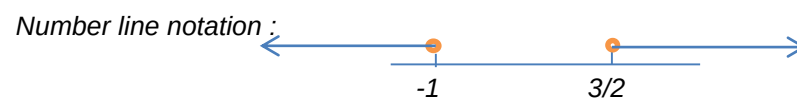
	$x < -1$	$-1 < x < 3/2$	$x > 3/2$
$(x + 1)$	-	+	+
$(2x - 3)$	-	-	+
$(x + 1)(2x - 3)$	+	-	+

Since we want positive values, the answers are $x < -1$ or $x > \frac{3}{2}$

We can write the solution in various form / notation :

$$\text{Inequality notation : } x < -1 \text{ or } x > \frac{3}{2}$$

$$\text{Interval notation : } (-\infty, -1) \cup (\frac{3}{2}, \infty)$$



Hence, the set of values of x for which f is increasing is

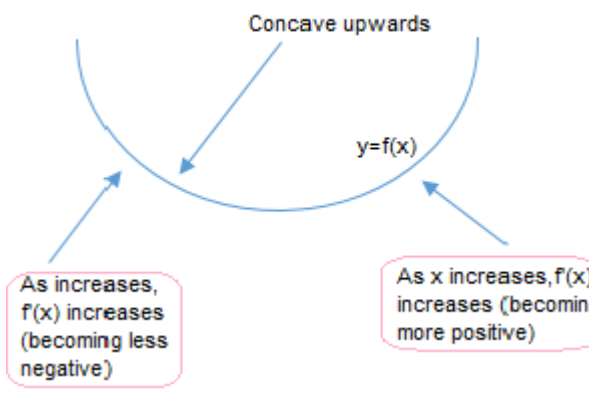
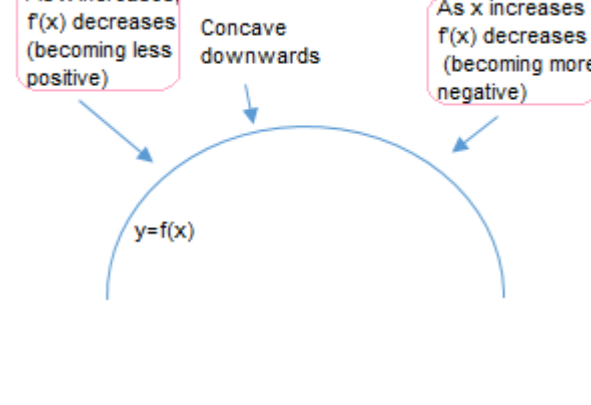
$$\{x : x < -1 \text{ or } x > \frac{3}{2}\}.$$

When f is decreasing, $f'(x) < 0$. Hence, the set of values of x for which f is decreasing is

$$\left\{x : -1 < x < \frac{3}{2}\right\}$$

Concavity of the graph of Function

- Let $y = f(x)$ be a function differentiable in the interval (a, b)

	
- The graph of $y = f(x)$ in the figure above is said to be concave upwards if all its tangents lie below it in the interval (a, b). - The graph of $y = f(x)$ is concave upwards if $f'(x)$ increases as x increases or the derivative of $f'(x)$, i.e. $f''(x) > 0$.	- The graph of $y = f(x)$ in the figure above is said to be concave downwards if all its tangents lie above it in the interval (a, b). - The graph of $y = f(x)$ is concave downwards if $f'(x)$ decreases as x increases or the derivative of $f'(x)$, i.e. $f''(x) < 0$.

Example:

Find the set of values of x for which the graph of the function

$$f(x) = 2x^3 - 3x^2 - 12x + 11$$

- a) is concave upwards,
b) is concave downwards.

Solution:

Differentiate $f(x)$ with respect to x .

$$f(x) = 2x^3 - 3x^2 - 12x + 11$$

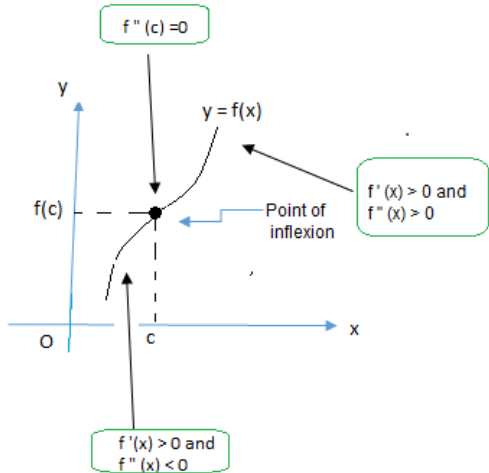
$$\text{first derivative, } f'(x) = 6x^2 - 6x - 12$$

$$\text{second derivative, } f''(x) = 12x - 6$$

a) When the graph of $f(x)$ is concave upwards,	a) When the graph of $f(x)$ is concave upwards,
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$f''(x) > 0$ $12x - 6 > 0$ $12x > 6$ $x > \frac{6}{12}$ $x > \frac{1}{2}$ Hence, the required set of values of x is $\left\{ x : x > \frac{1}{2} \right\}$	$f''(x) > 0$ $12x - 6 < 0$ $12x < 6$ $x < \frac{6}{12}$ $x < \frac{1}{2}$ Hence, the required set of values of x is $\left\{ x : x < \frac{1}{2} \right\}$
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Point of inflexion

	The point $P(c, f(c))$ on the curve $y = f(x)$ in figure is called the point of inflexion, such that the curve of $y = f(x)$ is continuous at point P and the curve changes from being concave downwards to concave upwards.
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